Day 18 7/11/26

206. Reverse linked list

Input 1→2→3→4→5→null
output 5→4→3→2→1→null

Approach: Three Pointers:

prev → previous node
curr → current node
next → store the next node temporarily

The key is to change the direction of curr.next without losing the remaining list.

Code

```
class Solution{
    public ListNode reverseList(ListNode head){
        ListNode prev = null;
        ListNod curr = head;

        while(curr != null){
        ListNode next = curr.next;
          curr.next = prev;
          prev = curr;
          curr = next;
        }
        return prev;
    }
}
```

Time Complexity: O(n)
Space Complexity: O(1)

Description | Accepted | Editorial | Solutions | Submissions

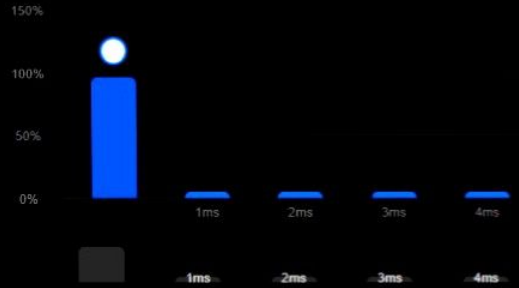
All Submissions

Runtime

0 ms | Beats **100.00%**

Memory

43.60 MB | Beats 88.87%



Code | Java

```
1  /**
```

Code

Java | Auto

```
10 */
11 class Solution {
12     public ListNode reverseList(ListNode head) {
13         ListNode prev = null;
14         ListNode temp = head;
15         while(temp != null){
16             ListNode next = temp.next;
17             temp.next=prev;
18             prev=temp;
19             temp=next;
20         }
21         return prev;
22
23     }
```

Saved

Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

head =

[1,2,3,4,5]